Software Engineering LAB-1

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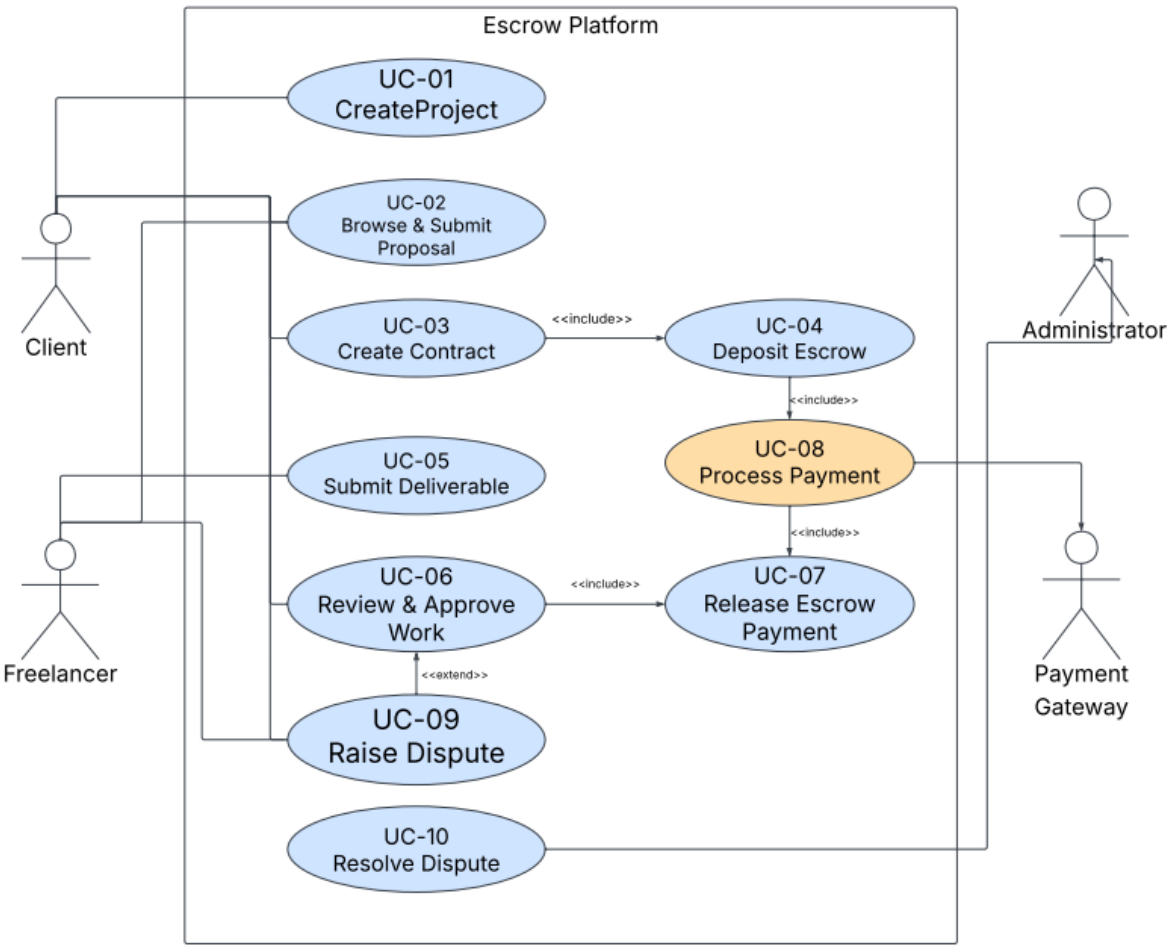
Section:G

1.Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall allow a client to create a project by specifying requirements, budget, deadline,and deliverables	High	When a client submits the "Create Project" form with all required fields completed, the project is saved and appears in the freelancer browse list within 5 seconds.	Core functionality; without a defined project, no proposals or contracts can exist.
FR-002	Functional	The system shall allow a freelancer to browse open projects and submit a proposal containing a price quote and timeline.	High	When a freelancer submits a proposal on an open project, the proposal is recorded with a timestamp and is visible to the client within 5 seconds.	Enables freelancer matching, which is the core marketplace mechanism.
FR-003	Functional	The system shall allow a client to select a freelancer's proposal and create a contract, and shall require the client to deposit the agreed amount into escrow before the freelancer can begin work.	High	A contract cannot transition to "Active" status, and the freelancer cannot mark work as started, until an escrow deposit equal to the contract amount is confirmed by the payment gateway.	Escrow protects both parties and is the central trust mechanism of the platform.
FR-004	Functional	The system shall allow a freelancer to submit completed deliverables, and allow the client to either approve the work or request revisions.	High	When the client selects "Approve," the contract status changes to "Completed" and escrow release is triggered; when the client selects "Request Revision," the freelancer	Directly satisfies the scenario's core deliverable-review workflow.

				receives a revision note and the contract stays "In Review."	
FR-005	Functional	The system shall allow either the client or the freelancer to raise a dispute on a contract, and shall allow an administrator to review the dispute and decide how the escrow amount is handled (release, refund, or split).	Medium	When a dispute is raised, the contract status changes to "Disputed" and escrow is frozen; when an administrator records a decision, escrow funds are distributed accordingly and the contract status changes to "Resolved" within 24 hours of the decision being recorded.	Provides a fair resolution path when the client and freelancer disagree, preventing loss of funds for either party.
NFR001	Nonfunc.	The system shall encrypt all escrow and payment transactions in transit and at rest, and shall never store raw payment card data on platform servers.	High	A security scan confirms TLS 1.2+ is enforced on all payment-related endpoints and that no PAN (card number) fields exist in the database schema.	A security scan confirms TLS 1.2+ is enforced on all payment-related endpoints and that no PAN (card number) fields exist in the database schema.
NFR002	Nonfunc.	The system shall release approved escrow payments to the freelancer's account within 2 minutes of client approval, 99% of the time.	Medium	A timing log of 100 sampled approvals shows escrow release completing in under 2 minutes for at least 99 of the 100 events.	Freelancers rely on prompt payment; slow release undermines trust in the escrow system.

2.UML Use-Case Diagram



3.Use-Case Flow Document

Use Case: Submit Work & Release Escrow Payment

Main Success Scenario

1.Freelancer selects “Submit Deliverable” on an active contract. System displays a form to upload files and enter a submission note.

2.Freelancer uploads the completed deliverable and submits it.

3.System validates the upload (file present, within size limits) and marks the contract status as “In Review.”

4.System notifies the Client that a deliverable is ready for review.

5.Client opens the submission and reviews the deliverable against the project requirements.

6.Client selects “Approve.”

7.System «includes» the Process Payment use case: it instructs the Payment Gateway to release the escrowed funds to the freelancer's account.

8.Payment Gateway confirms the transfer is authorized.

9.System updates the contract status to “Completed” and updates the freelancer's earnings balance.

10.A confirmation notification and receipt are sent to both the Client and the Freelancer.

11.Use case ends successfully.

Alternate Flow

6a. Client Requests Revision

- 6a1. Client selects “Request Revision” instead of “Approve” and enters revision notes.
- 6a2. System sets contract status back to “In Progress” and sends the revision notes to the Freelancer.
- 6a3. Freelancer revises the deliverable and resubmits, returning the flow to step 3.

6a4. If the Client and Freelancer cannot agree after repeated revisions, either party may raise a dispute